

Conjecture 00000003526: well-founded trees, their rank, and ordinal height

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Statement

Definition: Combinatorial principles of large ordinals: trees and partitions. Conjecture: The ordinal version of the tree lemma: the vanishing of descending chains of ordinal trees is well-foundedness, with the rank of well-foundedness the ordinal height.

Verdict. PROVED, for the two standard notions of tree named below. The claim is classical. We do not take it from a library theorem about trees. We write out the definitions of tree, branch, descending chain, rank and height, and we prove every equivalence from them. All of it is checked in Lean 4 with Mathlib.

1 Readings and definitions

Reading A (trees of finite sequences). We use the trees of descriptive set theory. Wikipedia, “Tree (descriptive set theory)”, says that a tree on X is a set of finite sequences “such that every prefix of a sequence in the collection also belongs to the collection”. It says that “a tree is a nonempty subset” of $X^{<\omega}$ with this closure property, and that it “shows that the empty sequence belongs to every tree”. A branch is an infinite sequence “each of whose finite prefixes belongs to” T , and “A tree that has no branches is called wellfounded; a tree with at least one branch is illfounded.” An *ordinal tree* is a tree on a set of ordinals. All our results hold for trees on an arbitrary set A , so they hold in particular for $A = \text{Ord}$.

Reading B (set-theoretic trees). Wikipedia, “Tree (set theory)”, says that “a tree is a partially ordered set” $(T, <)$ such that for each t the set $\{s : s < t\}$ “is well-ordered by the relation” $<$. The order type of that set “is called the height, rank, or level of” t , and the height of T “itself is the least ordinal greater than the height of each element of” T .

Definition 1 (Lean names in brackets). Let $T \subseteq A^{<\omega}$.

- T is a *tree* [IsTree] if $T \neq \emptyset$ and $t \in T, m < |t|$ imply $t \upharpoonright m \in T$.
- For $f : \mathbb{N} \rightarrow A$, $f \upharpoonright n = \langle f(0), \dots, f(n-1) \rangle$ [initSeg]. The sequence f is a *branch* [IsBranch] if $f \upharpoonright n \in T$ for all n . The tree T is *well-founded* [TreeWF] if it has no branch.
- *Strict extension* [Ext]: for nodes $s, t \in T$, $t \prec s$ iff s is a proper prefix of t . A *descending chain* is a sequence of nodes with $c_{n+1} \prec c_n$ for all n , that is, $c_0 \subsetneq c_1 \subsetneq \dots$ [HasDescChain]. This is the orientation in which “descending” means going away from the root.
- If $\prec$ is well-founded, the *rank* [nodeRank] is $\rho_T(s) = \sup\{\rho_T(t) + 1 : t \in T, t \prec s\}$, with $\sup \emptyset = 0$. This is Mathlib’s well-founded rank `Acc.rank`. The *rank of the tree* [treeRank] is $\rho(T) = \sup\{\rho_T(s) + 1 : s \in T\}$.

- A *ranking* [IsRanking] is a map $\varphi : T \rightarrow \text{Ord}$ with $t \prec s \Rightarrow \varphi(t) < \varphi(s)$. The *ordinal height* [ordHeight] is $\text{ht}(T) = \min\{\alpha : \exists \varphi \text{ a ranking with } \varphi(s) < \alpha \ \forall s\}$. This definition does not mention ρ .

For the name “ordinal height”: Huschenbett, Kartzow, Liu and Lohrey, *Tree-Automatic Well-Founded Trees* (arXiv:1201.5495), Section 2.2, write “Let us recall the (ordinal) rank of a well-founded partial order.” They define the rank of an element by recursion and then “the rank of” the order “(also called the ordinal height of” it) as $\sup\{\text{rank}(p) + 1\}$. This is exactly $\rho(T)$. Our $\text{ht}(T)$ is a second, independent notion of height, and Theorem 3 shows that the two agree.

Conventions. Ordinals live in Lean’s universe-polymorphic `Ordinal.{u}`. For a tree on A with A in `Type u`, ranks lie in `Ordinal.{u}`. For a tree on `Ordinal.{u}` itself, ranks lie in `Ordinal.{u+1}`, and α appears as `lift(α)`. In Lean, `nodeRank` is 0 on ill-founded trees and `ordHeight` is $\inf \emptyset = 0$ when no ranking exists. These junk values never enter a theorem, because every rank statement assumes well-foundedness. The step from “no descending chain” to “well-founded” uses dependent choice, which Lean’s `Classical.choice` provides. As Wikipedia, “Well-founded relation”, puts it: “Equivalently, assuming the axiom of dependent choice, a relation is well-founded when it contains no infinite descending chains, meaning there is no infinite sequence $x_0, x_1, x_2, \dots$ of elements of X such that $x_{n+1} R x_n$ for every natural number n .”

2 Reading A

Theorem 2 (`treeWF_iff_no_descChain, no_descChain_iff_wf`). *For a tree T , the following are equivalent: T has no branch; T has no descending chain; $\prec$ is well-founded on T .*

Proof. Let f be a branch. Then $c_n = f \upharpoonright n$ is a descending chain, because $f \upharpoonright (n+1) = f \upharpoonright n \frown f(n)$. Conversely, let (c_n) be a descending chain. By induction, $c_m \subseteq c_n$ for $m \leq n$. Since a proper prefix is strictly shorter, $|c_n| \geq n$. Put $f(k) = c_{k+1}(k)$. For $i < n$ we have $c_{i+1} \subseteq c_n$, so $f(i) = c_n(i)$. Hence $f \upharpoonright n = c_n \upharpoonright n$, which is a prefix of $c_n \in T$. By prefix closure (`IsTree.prefix_mem`) it lies in T , so f is a branch. The second equivalence is the descending-chain criterion (`wellFounded_iff_isEmpty_descending_chain`). $\square$

Theorem 3 (`nodeRank_eq_iSup_child, nodeRank_unique, nodeRank_le_of_ranking, wf_iff_exists_ranking`). *Let T be a tree. $\prec$ is well-founded if and only if T has a ranking. If T is well-founded, then: (a) $\rho_T(s) = \sup\{\rho_T(s \frown a) + 1 : s \frown a \in T\}$, the supremum over one-step extensions; (b) ρ_T is the only function satisfying the recursion (a); (c) $\rho_T(s) \leq \varphi(s)$ for every ranking φ , and ρ_T is itself a ranking.*

Proof. Suppose φ is a ranking. Then $\prec$ is a subrelation of the inverse image of $<$ on the ordinals under φ , so it is well-founded. Conversely, ρ_T is a ranking because $t \prec s$ implies $\rho_T(t) < \rho_T(s)$.

(a) Every one-step extension $s \frown a \in T$ satisfies $s \frown a \prec s$, which gives $\geq$. For $\leq$, take $t \prec s$. Then $t = s \frown a \frown w$, so $s \frown a$ is a prefix of t and lies in T . Hence $\rho_T(t) \leq \rho_T(s \frown a)$, with equality if $t = s \frown a$.

(b) Let φ satisfy the recursion. Well-founded induction along $\prec$, using (a), shows $\varphi(s) = \rho_T(s)$.

(c) By well-founded induction: if $\rho_T(t) \leq \varphi(t) < \varphi(s)$ for all $t \prec s$, then $\rho_T(s) = \sup_{t \prec s} (\rho_T(t) + 1) \leq \varphi(s)$. $\square$

Theorem 4 (`ordHeight_eq_treeRank, treeRank_eq_succ_root`). *If the tree T is well-founded, then $\text{ht}(T) = \rho(T) = \rho_T(\emptyset) + 1$.*

Proof. ρ_T is a ranking with values below $\rho(T)$, so $\text{ht}(T) \leq \rho(T)$. Let φ be a ranking bounded by α . By Theorem 2(c), $\rho_T(s) + 1 \leq \varphi(s) + 1 \leq \alpha$ for every s , so $\rho(T) \leq \alpha$. Every s extends $\emptyset$, so $\rho_T(s) \leq \rho_T(\emptyset)$, and therefore $\rho(T) = \rho_T(\emptyset) + 1$. $\square$

The Lean statement `tree_wf_rank_height` bundles Theorems 1–3 except the uniqueness of the one-step recursion (Theorem 2(b)), which is proved separately as `nodeRank_unique`.

Theorem 5 (`nodeRank_decTree`, `ordinal_tree_realization`). *For every ordinal α , let $D_\alpha = \{s : \alpha > s_0 > s_1 > \dots > s_k\}$ be the tree on the ordinals of strictly decreasing sequences below α (`decTree`). Then D_α is a well-founded tree, $\rho_{D_\alpha}(s)$ is the last entry of $\alpha \smallfrown s$, so $\rho_{D_\alpha}(\emptyset) = \alpha$, and $\text{ht}(D_\alpha) = \alpha + 1$.*

Proof. Let $b(s)$ be the last entry of $\alpha \smallfrown s$. We have $s \smallfrown a \in D_\alpha$ iff $s \in D_\alpha$ and $a < b(s)$. Along a proper extension, b strictly decreases, so b is a ranking and D_α is well-founded. Moreover $\sup\{a + 1 : a < b(s)\} = b(s)$, so b satisfies recursion (a), and Theorem 2(b) gives $\rho = b$. The rest follows from Theorem 3. $\square$

So every ordinal, and in particular every countable ordinal, is the root rank of a well-founded ordinal tree. Every successor ordinal is the height $\text{ht} = \rho(T)$ of such a tree. Under this convention the height of a nonempty tree is always a successor ordinal (Theorem 3).

3 Reading B

Theorem 6 (`setTree_wf_rank_height`). *Let $(P, <)$ be a set-theoretic tree [`IsSetTree`]. Then $<$ is well-founded, so descending chains vanish. The well-founded rank of each t [`setRank`] equals its height, the order type of $\{s : s < t\}$ [`ht`]. The height of P , the least ordinal above all heights [`setTreeHeight`], equals $\sup_t(\text{rank}(t) + 1)$.*

Proof. Suppose $f(n+1) < f(n)$ for all n . Then $(f(n+1))_n$ is a descending sequence in the well-order $\{s : s < f(0)\}$, which is impossible. For $s < t$, the map $z \mapsto z$ is an order isomorphism from $\{z : z < s\}$ onto the initial segment of $\{z : z < t\}$ below s , so $\text{ht}(s) = \text{typein}(s)$. By induction, $\text{rank}(t) = \sup_{s < t}(\text{typein}(s) + 1)$. Every typein is below $\text{type}\{z : z < t\}$, and every smaller ordinal is a typein , so this supremum equals $\text{ht}(t)$. The statement about the height of P follows. $\square$

Lean formalization and axiom audit

The file is `lean/Conjecture3526/Basic.lean` (383 lines, `import Mathlib` only). Its main theorems are `C3526.tree_wf_rank_height`, `C3526.ordinal_tree_realization` and `C3526.setTree_wf_rank_height`. The command `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for each of them. The file contains no `sorry` and no `native_decide`.

Sources

- Wikipedia, Tree (descriptive set theory), [https://en.wikipedia.org/wiki/Tree_\(descriptive_set_theory\)](https://en.wikipedia.org/wiki/Tree_(descriptive_set_theory)).
- Wikipedia, Tree (set theory), [https://en.wikipedia.org/wiki/Tree_\(set_theory\)](https://en.wikipedia.org/wiki/Tree_(set_theory)).
- Wikipedia, Well-founded relation, https://en.wikipedia.org/wiki/Well-founded_relation.

- M. Huschenbett, A. Kartzow, J. Liu, M. Lohrey, Tree-Automatic Well-Founded Trees, <https://arxiv.org/abs/1201.5495>.